

Qingfeng Zhang

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Education Background

PhD in Water Resources and Hydropower Engineering Wuhan University, School of Water Resources and Hydropower Engineering Supervisor: Prof. Jiyun Song	2024–Now
B.Eng. in Hydrology and Water Resources Engineering Northwest A&F University Supervisor: Prof. Yi Li – Ranked 1 st in the postgraduate exemption examination; exempted to Wuhan University for PhD. – Bachelor's thesis selected as one of the university's Top 100 Distinguished Papers.	2020–2024

Research Interests

- Urban climate dynamics, heat extremes, and land–atmosphere interactions
- Physics-informed spatiotemporal deep learning for high-resolution urban weather modeling and nowcasting
- Climate-just urban informatics and multi-objective optimization for heat-resilient cities

Journal Publications

(*: Corresponding author)

Published Journal Papers:

1. Zhong Q, Song J*, Wang X, Zhang J, **Zhang Q**, Li Y. 2026. Modelling of lake–land breeze circulation in a diurnal cycle. *Building and Environment*, 293: 114319. [doi:10.1016/j.buildenv.2026.114319](https://doi.org/10.1016/j.buildenv.2026.114319).
2. **Zhang Q**, Li Y*, Hu Q, Yao N, Song X, Liu F, Pulatov B, Meng Q, Feng P. 2022. Impacts of Different Socioeconomic Development Levels on Extremely Wet/Dry Events in Mainland China. *Water*, 14(23): 3950. [doi:10.3390/w14233950](https://doi.org/10.3390/w14233950).

Manuscripts Submitted / Under Review:

1. **Zhang Q**, Song J*, Liu H. Physics-informed spatio-temporal graph neural networks for fine-resolution multi-variable urban weather nowcasting from sparse station observations. Submitted to *Atmospheric Chemistry and Physics*.
2. **Zhang Q**, Song J*, Zhou Y, Zhang J, Zhou Z. Physics-guided deep learning resolves five-metre urban thermal stress for climate-just delivery routing. Submitted to *npj Urban Sustainability*.
3. **Zhang Q**, Song J*, Li Y, Zhou Y. Heatwave risk reshaped by urban expansion-clustering nexus. Under review at *Science Advances*.

Conference Presentations

1. **Zhang Q**, Song J. *An Inductive Spatio-Temporal Graph Network for Fine-Resolution Urban Weather Nowcasting Integrating Multi-Source Data and Physical Constraints*. EGU General Assembly 2026, Vienna, Austria, May 3–8, 2026. EGU26-2288. <https://doi.org/10.5194/egusphere-egu26-2288>. (Oral)
2. Song J, **Zhang Q**. *AI-uWRF: A Physics-Informed Spatiotemporal Diffusion Transformer for High-Resolution Urban Weather Emulation*. EGU General Assembly 2026, Vienna, Austria, May 3–8, 2026. EGU26-2266. <https://doi.org/10.5194/egusphere-egu26-2266>. (Oral)
3. **Zhang Q**, Song J. *Urban Climate Modeling with AI-Hybrid Approaches*. Hong Kong Workshop, Hong Kong, China, November 26–28, 2025. (Oral)
4. **Zhang Q**, Song J. *A Hybrid Modeling Framework for High-Resolution Wet-Heat Mapping in Metropolitan Areas: Integrating Physical Simulation and Deep Learning*. The 4th IAHR-YPN PhD Forum, Dalian, China, September 26–28, 2025. (Oral)
5. **Zhang Q**, Song J. *A Hybrid Modeling Framework for High-Resolution Wet-Heat Mapping in Metropolitan Areas: Integrating Physical Simulation and Deep Learning*. The 22nd China Water Forum, Xi'an, China, September 12–14, 2025. (Oral)
6. Song J, **Zhang Q**. *The Role of Dynamic Urban Expansion in Shaping Thermal Environments: Evidence from China's 2000 Cities in Past Decades*. The 22nd Annual Meeting of the Asia Oceania Geosciences Society (AOGS), Singapore, July 28–August 1, 2025. HS41-A005. (Oral)
7. **Zhang Q**, Song J. *A Novel Hybrid Modeling Framework for Urban Heat Stress: Combining Physical Simulations and CNN-Transformer Networks in Wuhan*. The 22nd Annual Meeting of the Asia Oceania Geosciences Society (AOGS), Singapore, July 28–August 1, 2025. HS41-A006. (Poster)
8. Song J, **Zhang Q**. *A Hybrid Modeling Framework for High-Resolution Humid Heat Mapping in Metropolitan Wuhan: Integrating Physical Simulations and Deep Learning*. The 12th International Conference on Urban Climate (ICUC12), Rotterdam, The Netherlands, July 7–11, 2025. ICUC12-201. <https://doi.org/10.5194/icuc12-201>. (Oral)
9. **Zhang Q**, Song J. *The role of dynamic urban expansion in shaping thermal environments: Evidence from China's 2000 cities in past decades*. The 12th International Conference on Urban Climate (ICUC12), Rotterdam, The Netherlands, July 7–11, 2025. ICUC12-171. <https://doi.org/10.5194/icuc12-171>. (Oral)
10. Song J, **Zhang Q**. *Impacts of Dynamic Urban Expansion on Heatwaves in Urban Agglomerations: Empirical Analysis of Nearly 2,000 Cities in China, 1990–2022*. The 3rd Sustainable City Workshop, Wuhan, China, May 23–25, 2025. (Oral)
11. **Zhang Q**, Song J. *Impacts of Dynamic Urban Expansion on Heatwaves in Urban Agglomerations: Empirical Analysis of Nearly 2,000 Cities in China, 1990–2022*. The 10th Youth Geoscience Forum, Hefei, China, May 9–12, 2025. (Oral)

Research & Project Experience

- 1. PhD Research: Physics-Informed Deep Learning for Urban Weather Nowcasting and Emulation** 2024–Now
Wuhan University, School of Water Resources and Hydropower Engineering
 - **Causal ML for urban expansion–heatwave nexus.** Used double machine learning to decouple urbanization-induced thermal penalties from climate variability across 1,690 Chinese cities (1990–2022); revealed a dichotomy between internal 3D densification (local intensity/onset) and intercity coalescence (regional frequency/duration), and identified a ~ 50 km spatial threshold for mega urban heat dome formation. Under review at *Science Advances*.
 - **HyperClim-Former: 5-metre thermal stress mapping and climate-just routing.** Developed a physics-guided deep learning framework coupling mesoscale forcings with 3D urban morphology for 5 m, 10-min UTCI reconstruction (RMSE = 1.00°C); integrated with agent-based courier simulations and multi-objective Pareto optimization to reduce heat stress by 21.1% with only 4.3 min added delay per shift. Submitted to *npj Urban Sustainability*.
 - **Physics-informed STGNN for station-to-grid urban weather nowcasting.** Designed five physically motivated graph topologies (proximity, thermal similarity, wind advection, terrain, LCZ) for 333 m station-to-grid downscaling; implemented WRF-to-observation transfer learning with thermodynamic constraints, reducing WRF biases by 25–67% while retaining 92.7% spatial heterogeneity. Submitted to *Atmospheric Chemistry and Physics*.
- 2. Cumulative Effects of Hydrology and Ecology in the Yellow River Basin** 2022–2023
Project Lead, Northwest A&F University
 - Revealed the hydrology–ecosystem coupling effect in the Yellow River Basin, established a large-scale cumulative effect model, and developed remote sensing data pipelines to support the research findings.
- 3. Development and Design of Robot Visual Aid System** 2021–2022
Core Member, Northwest A&F University
 - Programmed and debugged OpenCV-based real-time object detection under Linux for RoboMaster college league robotics and agricultural picking robots.
- 4. Regional Water and Food Big Data Platform** 2021–2022
Core Member, Northwest A&F University
 - Analyzed regional water consumption patterns and database architecture to map spatiotemporal agricultural water use efficiency.

Competition Awards

1. 2025 China Graduate Smart City Technology and Creative Design Competition, Second Prize, Party Committee Graduate Work Department of Wuhan University, 2025.09.
2. The 4th IAHR-YPN PhD Forum, Report Innovation Award, Dalian, China, 2025.09.
3. The 10th Youth Geoscience Forum, Excellent Student Presentation Award, Hefei, China, 2025.05.
4. The Eighth National College Students' Water Resources Innovation Design Competition, National Level Grand Prize, China Water Resources Education Association, 2023.08.
5. The 14th National Student Mathematics Competition (Non-Mathematics), Provincial First Prize, Chinese Mathematical Society, 2023.01.
6. China Engineering Robotics Competition and International Open Competition 2022, National First Prize and Third Runner-up, Chinese Society of Artificial Intelligence, 2022.12.
7. 2022 National College Robotics Competition (Robomasters) College League Competition and Sectional Competition, Third Prize National Level, Chinese Society of Artificial Intelligence, 2022.07.
8. National University Students' Surveying and Mapping Skills Competition (Non-specialized Group of Surveying and Mapping Programming Competition), 2022, National Second Prize, China Association of Surveying and Mapping, 2022.05.
9. 2022 Mathematical Contest In Modeling, International Honorable Mention, the Consortium for Mathematics and Its Application, 2022.02.
10. Huawei Cup 2nd National College Students' Innovation Design Competition in Agricultural Water Conservancy Engineering and Related Specialties, National Level Grand Prize, China Agricultural Engineering Society, 2021.11.
11. 2021 China Engineering Robotics Competition and International Open Competition, National Second Prize, Chinese Society of Artificial Intelligence, 2021.11.
12. The 12th Blue Bridge Cup National Software and Information Technology Professionals Competition, Provincial Second Prize, Talent Exchange Center of Ministry of Industry and Information Technology, 2021.05.

Skills

- **HPC & Numerical Weather Prediction:** WRF and WRF-UCM (multi-domain nesting, urban canopy schemes) on HPC platforms; MPI parallel computing; large-scale NetCDF data processing.
- **AI & Spatiotemporal Deep Learning:** Spatiotemporal graph neural networks, diffusion models, Transformers, CNNs; physics-informed neural networks; causal inference; transfer learning; SHAP.
- **Scientific Programming:** Python (PyTorch, xarray, pandas, scikit-learn), C++, R, MATLAB; distributed and mixed-precision training; agent-based modeling; Linux/HPC environments.
- **Geospatial & Microclimate Analysis:** ArcGIS, QGIS, UMEP, ENVI-met, CAD, Origin; LiDAR, satellite, and mobile sensor fusion; local climate zone classification; thermal comfort indices.
- **Languages:** English (CET-6: 536, CET-4: 550), Chinese (Native).